

Classifier transfer learning from LLM 情緒類別分類器(遷移學習)

This is a comprehensive notebook and tutorial on how to fine tune the `qwen-0.5b` classification model

Fine-tuning Qwen-0.5B (a smaller model) with LoRA (Low-Rank Adaptation) is an efficient approach that requires less computational power.

It got a classification accuracy of 0.93.

Transformers 變形金剛(變壓器)模型

Bert and GPT are parts of Transformer

Encoders and Decoders As mentioned, there are encoders and decoders. BERT uses encoders only, GTP uses decoders only. Both options understand language including syntax and semantics. Especially the next generation of large language models like GPT with billions of parameters do this very well.

The two models focus on different scenarios. However, since the field of foundation models is evolving, the differentiation is often fuzzier.

BERT (encoder): classification (e.g., sentiment), questions and answers, summarization, named entity recognition GPT (decoder): translation, generation (e.g., stories) The outputs of the core models are different:

BERT (encoder): Embeddings representing words with attention information in a certain context
GPT (decoder): Next words with probabilities Both models are pretrained and can be reused without intensive training. Some of them are available as open source and can be downloaded from communities like Hugging Face, others are commercial. Reuse is important, since trainings are often very resource intensive and expensive which few companies can afford.

The pretrained models can be extended and customized for different domains and specific tasks. Layers can sometimes be reused without modifications and more layers are added on top. If layers need to be modified, the new training is more expensive. The technique to customize these models is called Transfer Learning, since the same generic model can easily be transferred to other domains.

[Source](#)

Encoder-Decoder Transformers

Attention Layer

Colab GPU

設定執行階段 變更執行階段類型 硬體加速器 選取 GPU

```
!nvidia-smi

Sat May 24 15:27:27 2025
+-----+
+-----+
| NVIDIA-SMI 550.54.15                  Driver Version: 550.54.15
| CUDA Version: 12.4                    |
+-----+-----+
+-----+-----+
| GPU  Name                               Persistence-M | Bus-Id          Disp.A |
| Volatile Uncorr. ECC |
| Fan  Temp  Perf              Pwr:Usage/Cap |      Memory-Usage |
| GPU-Util  Compute M. |
| MIG M. |
+-----+-----+
=====+=====
=====+=====
|    0   Tesla T4                               Off |  00000000:00:04.0 Off |
| 0 |
| N/A    69C    P8              10W /   70W |      0MiB /  15360MiB |
| 0%      Default |
| N/A |
+-----+-----+
+-----+-----+
+-----+-----+
| Processes:
|
| GPU   GI    CI          PID    Type    Process name
GPU Memory |
|      ID    ID
Usage      |
```

```

|
=====
=====|
|   No running processes found
|
+-----+
-----+

# 列出GPU與CPU資源
from tensorflow.python.client import device_lib
print(device_lib.list_local_devices())

[name: "/device:CPU:0"
 device_type: "CPU"
 memory_limit: 268435456
 locality {
 }
 incarnation: 14346530960944514404
 xla_global_id: -1
 , name: "/device:GPU:0"
 device_type: "GPU"
 memory_limit: 14619377664
 locality {
   bus_id: 1
   links {
 }
 }
 incarnation: 14489212002461153143
 physical_device_desc: "device: 0, name: Tesla T4, pci bus id:
 0000:00:04.0, compute capability: 7.5"
 xla_global_id: 416903419
]

# Specify the GPU device to use
#import os
#os.environ["CUDA_VISIBLE_DEVICES"] = "0"

```

Ubuntu 版本資訊

```

!lsb_release -a

No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.4 LTS
Release:       22.04
Codename:      jammy

```

Insatll packages

```
transformers: For model handling.  
peft: For LoRA integration.  
datasets: For dataset handling.  
accelerate: For distributed training.  
bitsandbytes: For memory-efficient training (optional but useful  
for Qwen).
```

```
#!pip install transformers peft datasets accelerate bitsandbytes
```

```
!pip install evaluate
```

```
Collecting evaluate
```

```
  Downloading evaluate-0.4.3-py3-none-any.whl.metadata (9.2 kB)
```

```
Requirement already satisfied: datasets>=2.0.0 in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (2.14.4)
```

```
Requirement already satisfied: numpy>=1.17 in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (2.0.2)
```

```
Requirement already satisfied: dill in /usr/local/lib/python3.11/dist-  
packages (from evaluate) (0.3.7)
```

```
Requirement already satisfied: pandas in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (2.2.2)
```

```
Requirement already satisfied: requests>=2.19.0 in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (2.32.3)
```

```
Requirement already satisfied: tqdm>=4.62.1 in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (4.67.1)
```

```
Requirement already satisfied: xxhash in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (3.5.0)
```

```
Requirement already satisfied: multiprocessing in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (0.70.15)
```

```
Requirement already satisfied: fsspec>=2021.05.0 in  
/usr/local/lib/python3.11/dist-packages (from fsspec[http]>=2021.05.0-  
>evaluate) (2025.3.2)
```

```
Requirement already satisfied: huggingface-hub>=0.7.0 in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (0.31.2)
```

```
Requirement already satisfied: packaging in  
/usr/local/lib/python3.11/dist-packages (from evaluate) (24.2)
```

```
Requirement already satisfied: pyarrow>=8.0.0 in  
/usr/local/lib/python3.11/dist-packages (from datasets>=2.0.0-  
>evaluate) (18.1.0)
```

```
Requirement already satisfied: aiohttp in  
/usr/local/lib/python3.11/dist-packages (from datasets>=2.0.0-  
>evaluate) (3.11.15)
```

```
Requirement already satisfied: pyyaml>=5.1 in  
/usr/local/lib/python3.11/dist-packages (from datasets>=2.0.0-  
>evaluate) (6.0.2)
```

```
Requirement already satisfied: filelock in  
/usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.7.0-  
>evaluate) (3.18.0)
```

Requirement already satisfied: typing-extensions>=3.7.4.3 in
/usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.7.0-
>evaluate) (4.13.2)

Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.19.0-
>evaluate) (3.4.2)

Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.19.0-
>evaluate) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.19.0-
>evaluate) (2.4.0)

Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.19.0-
>evaluate) (2025.4.26)

Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.11/dist-packages (from pandas->evaluate)
(2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.11/dist-packages (from pandas->evaluate)
(2025.2)

Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.11/dist-packages (from pandas->evaluate)
(2025.2)

Requirement already satisfied: aiohappyeyeballs>=2.3.0 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (2.6.1)

Requirement already satisfied: aiosignal>=1.1.2 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (1.3.2)

Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (25.3.0)

Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (1.6.0)

Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (6.4.3)

Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (0.3.1)

Requirement already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.11/dist-packages (from aiohttp-
>datasets>=2.0.0->evaluate) (1.20.0)

Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2-
>pandas->evaluate) (1.17.0)

Downloading evaluate-0.4.3-py3-none-any.whl (84 kB)

0:00:00

84.0/84.0 kB 2.9 MB/s eta

掛載雲端硬碟

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

切換工作目錄到雲端硬碟目錄下(取用資料比較方便，可以用相對路徑)

你要改成你自己雲端硬碟目錄的路徑，可複製路徑並貼上，不要用鍵盤輸入!

```
# cd to your folder
# cd 命令的前面一行不要加上說明文字，否則 colab 的 cd 會認不得指令

cd /content/drive/MyDrive/Colab Notebooks

/content/drive/MyDrive/Colab Notebooks

ls -l

total 14860
-rw----- 1 root root 312580 May 18 16:44 '「1-Training model using Transformers Trainer.ipynb」的副本'
-rw----- 1 root root 49197 May 24 15:27 '「1-改尾分類微調情緒分類 Qwen0.5b-v4-5epochs-acc0.93.ipynb」的副本'
-rw----- 1 root root 47656 May 19 05:34 '「2-sentiment classification model usage.ipynb」的副本'
drwx----- 2 root root 4096 May 13 02:50 best-model-v1/
drwx----- 2 root root 4096 May 13 02:35 checkpoints_v1/
drwx----- 2 root root 4096 May 20 04:17 checkpoints_v4/
-rw----- 1 root root 14071136 May 13 02:33 dataset_reviews.csv
-rw----- 1 root root 324 Oct 26 2023 Untitled0.ipynb
-rw----- 1 root root 11527 Feb 13 06:56 Untitled10.ipynb
-rw----- 1 root root 49223 Mar 9 15:14 Untitled11.ipynb
-rw----- 1 root root 4504 Mar 3 02:12 Untitled12.ipynb
-rw----- 1 root root 258 Mar 9 14:59 Untitled13.ipynb
-rw----- 1 root root 91763 Mar 10 03:23 Untitled14.ipynb
-rw----- 1 root root 38925 Mar 24 01:59 Untitled15.ipynb
-rw----- 1 root root 40608 Mar 24 12:31 Untitled16.ipynb
-rw----- 1 root root 495 Apr 28 01:40 Untitled17.ipynb
-rw----- 1 root root 48418 Apr 28 02:27 Untitled18.ipynb
```

```
-rw----- 1 root root      324 Oct 26  2023 Untitled1.ipynb
-rw----- 1 root root  257049 Jan  4  2024 Untitled2.ipynb
-rw----- 1 root root   14151 Jul 31  2024 Untitled3.ipynb
-rw----- 1 root root    2974 Sep  3  2024 Untitled4.ipynb
-rw----- 1 root root    2951 Aug 13  2024 Untitled5.ipynb
-rw----- 1 root root    816 Sep 25  2024 Untitled6.ipynb
-rw----- 1 root root    675 Oct 11  2024 Untitled7.ipynb
-rw----- 1 root root    324 Jan 10 02:02 Untitled8.ipynb
-rw----- 1 root root   9601 Feb 20 05:57 Untitled9.ipynb
-rw----- 1 root root   8659 Sep 15  2023 初版.ipynb
-rw----- 1 root root  133783 Mar 24 02:19 '「歡迎使用 Colaboratory」的副本'
```

```
import torch
import datasets
import pandas as pd
import evaluate
import numpy as np

# Load Huggingface transformers
from transformers import AutoTokenizer,
AutoModelForSequenceClassification, AutoModelForCausalLM
from transformers import TrainingArguments, Trainer
from transformers import BertTokenizer, BertTokenizerFast,
BertForSequenceClassification
from sklearn import metrics
import torch
import evaluate
import datasets

# Setting device on GPU if available, else CPU
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print('Using device:', device)

Using device: cuda
```

Read data

簡體中文資料，資料集來自於網路，轉換成繁體中文

```
df = pd.read_csv('./dataset_reviews.csv', sep='|')  
df  
  
{"summary": "{\n  \"name\": \"df\", \n  \"rows\": 80442, \n  \"fields\": [\n    {\n      \"column\": \"text\", \n      \"properties\": {\n        \"dtype\": \"string\", \n        \"num_unique_values\": 77059, \n        \"samples\": [\n          \"?\\u597d\\u8a55\\uff01?\\u597d\\u597d\\
```

```

u8a55??\u597d?\u597d\u8a55?\u597d\u8a55??\u597d?\u597d\u
u8a55??\u597d\u8a55?\u597d\u8a55?\u597d\u8a55??\u597d?\u597d\u
u8a55??\u597d?\u597d\u8a55\u0001",\n          "\u5abd\u7684\u
uff0c\u8cb7\u5c0f\u4e86\u9000\u56de\u53bb\u63db\u90fd\u4e00\u
u500b\u6708\u591a\u4e86\u9084\u6c92\u6709\u8a18\u56de\u4f86\u
u7d66\u6211\u0001\u60f3\u9000\u53c8\u9000\u4e0d\u4e86\u
uff0c***\u4e86\u0001\u7576\u8cb7\u500b\u6559\u8a13\u5427\u
uff0c\u4ee5\u5f8c\u90fd\u4e0d\u6703\u8cb7\u9019\u500b\u724c\u
u5b50\u7684\u8863\u670d\u0001\u6700\u4e0d\u958b\u5fc3\u7684\u
u4e00\u6b21\u8cfc\u7269\u0001\u5dee\u8a55\u5dee\u8a55\u5dee\u
u8a55\u5dee\u8a55\u7d55\u5c0d\u7684\u5dee\u8a55",\n
"\u8932\u5b50\u5f88\u597d\u770b\u5f88\u5408\u8eab\u0001\u
u54a8\u8a62\u6642\u5e97\u4e3b\u4e5f\u5f88\u71b1\u5fc3\u0001\u
u5f88\u4e0d\u932f\u7684\u4e00\u6b21\u8cfc\u7269\u0001\u4e94\u
u661f"\n          ],\n          "semantic_type": "",\n
"description": ""\n          }\n          },\n          {\n          "column":
"label",\n          "properties": {\n          "dtype": "number",\n
"std": 0,\n          "min": 0,\n          "max": 1,\n
"num_unique_values": 2,\n          "samples": [\n          0,\n
1\n          ],\n          "semantic_type": "",\n
"description": ""\n          }\n          }\n          ]\n
n"},"type":"dataframe","variable_name":"df"}

```

```
df.dtypes
```

```

text      object
label      int64
dtype: object

```

Convert the format of y

Convert the format of y from int to LongTensor

Convert label using one-hot representation 輸出資料格式 one-hot 轉換

轉成用 2 個節點表達兩類

```

類別 0: [1 0]
類別 1: [0 1]

```

如果是 3 個類別用 3 個節點表之：

```

類別 0: [1 0 0]
類別 1: [0 1 0]
類別 2: [0 0 1]

```

負面情緒 Negative 0 --> [1 0]

正面情緒 Positive 1 --> [0 1]

Easy Representation 0,1,2... (內部會自動轉換為 one-hot)

負面情緒 Negative 0 --> [0]

正面情緒 Positive 1 --> [1]

如果是 3 個類別

中立情緒 Neutral 2 --> [2]

```
# Map labels to integers
categories=['負面', '正面']
```

```
label_to_id = { cate : i for i, cate in enumerate(categories)}
```

```
label_to_id
```

```
{'負面': 0, '正面': 1}
```

```
id_to_label = { i : cate for i, cate in enumerate(categories)}
```

```
id_to_label
```

```
{0: '負面', 1: '正面'}
```

Sample some examples for demonstration

```
df = df.sample(10000)
```

Conver pandas dataframe to Huggingface Dataset

```
dataset = datasets.Dataset.from_pandas(df, preserve_index=False)
# eval_data = Dataset.from_pandas(X_eval)
```

```
dataset
```

```
Dataset({
  features: ['text', 'label'],
  num_rows: 10000
})
```

```
dataset[0]
```

```
{'text': '當時活動價買的挺便宜，真心不知道是不是假貨！！洗了一段時間發現頭髮好油好油！！真的好油！！一開始以為是護髮素用多了，後來發現不用護髮素都油的不行，自從用了惠潤的就這樣！現在頭髮1天洗一次都油的不行，真心覺得是假貨！！',  
  'label': 0}
```

```
dataset.to_pandas().head(5)
```

```
{  
  "summary": {  
    "name": "dataset",  
    "rows": 5,  
    "fields": [  
      {  
        "column": "text",  
        "properties": {  
          "dtype": "string",  
          "num_unique_values": 5,  
          "samples": [  
            "\u623f\u9593\u5f88\u5c0f\u0c\u8207\u884c\u653f\u9593\u7684\u540d\u5b57\u4e0d\u7b26\u3002\u53e6\u9019\u5bb6\u9152\u5e97\u7684\u6309\u6469\u53ef\u5343\u842c\u4e0d\u80fd\u53eb\u0c\u8d85\u7d1a\u9ed1\u0c\u7d50\u5e33\u6642\u5c45\u7136\u6703\u6709\u6253\u624b\u4e0a\u4f86\u0c\u8acb\u5927\u5bb6\u6ce8\u610f\u0c\u6c7a\u4e0d\u6703\u518d\u5165\u4f4f\u6b64\u9152\u5e97\u3002",  
            "\u7576\u7576\u5c31\u56e0\u70ba\u9019\u500b\u76e4\u7684\u9000\u63db\u8ca8\u8b93\u6211\u5fb9\u5e95\u8a8d\u8b58\u5230\u7576\u7576\u53ea\u8cb7\u4e0d\u9000\u63db\u81f3\u5c11\u4e0d\u771f\u5fc3\u9000\u63db\u8b93\u6211\u5011\u77ad\u89e3\u5230\u7576\u7576\u4e0d\u8aa0\u4fe1\u6211\u7684\u597d\u5e7e\u500b\u670b\u53cb\u5747\u6709\u540c\u611f\u0c",  
            "\u670d\u52d9\u5f88\u5dee\u0c\u623f\u9593\u6c92\u6709\u7a97\u6236\u0c\u544a\u8a34\u6211\u300c\u4e0d\u4fdd\u8b49\u63d0\u4f9b\u6709\u7a97\u6236\u7684\u623f\u9593\u300d\u0c\u6240\u4ee5\u5b89\u6392\u4f4f\u9032\u4e86\u4e00\u500b\u50c9\u6dle\u4e00\u6a23\u7684\u623f\u9593\u4e0d\u8aaa\u0c\u800c\u4e14\u4e2d\u9593\u9084\u6709\u6211\u4e00\u6839\u5f88\u602a\u7684\u5927\u67f1\u5b50\u0c\u771f\u662f\u5947\u8469\u2026\u2026\u0c\u4e0b\u6b21\u7d55\u5c0d\u4e0d\u4f4f\u4e86\u0c",  
          ],  
          "semantic_type": "",  
          "description": ""  
        },  
        "column": "label",  
        "properties": {  
          "dtype": "number",  
          "std": 0,  
          "min": 0,  
          "max": 1,  
          "num_unique_values": 2,  
          "samples": [  
            0,  
            1  
          ],  
          "semantic_type": "",  
          "description": ""  
        }  
      ]  
    },  
    "type": "dataframe"  
  }
```

Load Tokenizer

```
# model_id = "google/gemma-3-1b"  
# model_id = "Qwen/Qwen2.5-0.5B"  
model_id = "Qwen/Qwen2.5-0.5B-instruct"  
  
tokenizer = AutoTokenizer.from_pretrained(model_id)
```

```

/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(

{"model_id": "1c143247e25b47c59d1df116f5ac9cfd", "version_major": 2, "version_minor": 0}

{"model_id": "a562eec4f757418fbe43db77dc9a6489", "version_major": 2, "version_minor": 0}

{"model_id": "287e571367f14383803ff2e48e6a4cd0", "version_major": 2, "version_minor": 0}

{"model_id": "4a3b99cdc1bc4eb8914c2c936ec620bc", "version_major": 2, "version_minor": 0}

# 必須在Huggingface 註冊, 取得API token 才能下載模型
#access_token = "???"
#tokenizer = AutoTokenizer.from_pretrained(model_id,
token=access_token)

```

Tokenize text

```

def tokenize_function(example):
    return tokenizer(
        example["text"],
        #padding="max_length",
        max_length=512,
        truncation=True,
    )

tokenized_dataset = dataset.map(tokenize_function, batched=True)

{"model_id": "404f455b3afd4a349a08f40a831ad8bb", "version_major": 2, "version_minor": 0}

tokenized_dataset

Dataset({
  features: ['text', 'label', 'input_ids', 'attention_mask'],
  num_rows: 10000
})

```

```
# tokenized_dataset[0]
```

Split dataset for training and testing

Split dataset: Train, Test (Val)

Training set: 訓練資料集 --> 給模型讀進去訓練

Test set: 測試資料集 --> 驗證或測試模型的準確度

Split dataset into 90% for training and 10% for testing

```
tokenized_dataset = tokenized_dataset.train_test_split(test_size=0.05,
seed=1234)
train_data = tokenized_dataset["train"]
test_data = tokenized_dataset["test"]

train_data
Dataset({
  features: ['text', 'label', 'input_ids', 'attention_mask'],
  num_rows: 9500
})

print(test_data)
Dataset({
  features: ['text', 'label', 'input_ids', 'attention_mask'],
  num_rows: 500
})

train_data[0]
{'text': '賓館服務非常好，環境也很好，但在賓館吃飯太貴了，地角很偏，交通很不方便。',
 'label': 1,
 'input_ids': [117993,
110057,
100831,
108014,
3837,
108131,
74763,
101243,
3837,
104965,
117993,
```

[illegible]

定義 Model

這裡的做法有簡單的版本也有複雜的版本

簡單版:

```
model = AutoModelForSequenceClassification.from_pretrained(
    model_id, # 模型名稱
    num_labels=len(categories), # Number of output labels
)
```

複雜版:

```
from transformers import Qwen2Model, Trainer, TrainingArguments,
AutoTokenizer, AutoModel
from torch import nn
from transformers.modeling_outputs import SequenceClassifierOutput
import os

import torch
import os
import torch.nn.functional as F
from torch import nn

class QwenForClassifier(nn.Module):
    def __init__(self, base_model, hidden_size, num_labels):
        super(QwenForClassifier, self).__init__()
        # 凍結base model 的參數
        self.base_model = base_model

        for param in self.base_model.parameters():
            param.requires_grad = False

        # 注意力池化機制
        self.attention_pooler = nn.Sequential(
            nn.Linear(hidden_size, 128),
            nn.Tanh(),
            nn.Linear(128, 1)
        )

        # 多層融合權重(最後4層)
        self.layer_weights = nn.Parameter(torch.ones(4) / 4)

        # 增強型分類器
        self.classifier = nn.Sequential(
            nn.Linear(hidden_size, 256),
            nn.LayerNorm(256),
            nn.GELU(),
            nn.Dropout(0.2),

            nn.Linear(256, 128),
            nn.LayerNorm(128),
            nn.GELU(),
            nn.Dropout(0.1),
```

```

        nn.Linear(128, num_labels)
    )

    # 保存配置
    self.config = base_model.config
    self.config.num_labels = num_labels

def forward(self, input_ids, attention_mask=None, labels=None):
    # 獲取所有隱藏層狀態
    outputs = self.base_model(
        input_ids=input_ids,
        attention_mask=attention_mask,
        output_hidden_states=True
    )

    # 獲取最後4層隱藏狀態
    hidden_states = outputs.hidden_states
    if hidden_states is None:
        # 如果模型沒有返回hidden_states, 使用last_hidden_state
        last_hidden = outputs.last_hidden_state
        sequence_output = last_hidden
    else:
        # 融合最後4層(或可用層數)
        last_layers = hidden_states[-4:] if len(hidden_states) >=
4 else hidden_states[1:]
        layer_weights =
F.softmax(self.layer_weights[:len(last_layers)], dim=0)

        # 加權融合多層特徵
        sequence_output = torch.zeros_like(last_layers[0])
        for i, layer in enumerate(last_layers):
            sequence_output += layer_weights[i].unsqueeze(-
1).unsqueeze(-1) * layer

    # 注意力池化
    attention_scores = self.attention_pooler(sequence_output)
    attention_probs = F.softmax(attention_scores, dim=1)
    context_vector = torch.matmul(attention_probs.transpose(-1, -
2), sequence_output).squeeze(1)

    # 也計算平均池化向量
    mean_pooled = torch.mean(sequence_output, dim=1)

    # 結合注意力池化和平均池化(殘差連接)
    combined_repr = context_vector + mean_pooled

    # 分類預測
    logits = self.classifier(combined_repr)

```

```

# 計算損失
loss = None
if labels is not None:
    loss_fct = nn.CrossEntropyLoss()
    loss = loss_fct(logits, labels)

return {"loss": loss, "logits": logits}

def save_model(self, output_dir=None):
    """ 保存分類器權重和配置 """
    os.makedirs(output_dir, exist_ok=True)

    # 保存分類器權重
    classifier_path = os.path.join(output_dir,
    "classifier_weights.pt")
    model_dict = {
        'classifier': self.classifier.state_dict(),
        'attention_pooler': self.attention_pooler.state_dict(),
        'layer_weights': self.layer_weights,
        'config': {
            'num_labels': self.config.num_labels,
            'hidden_size': self.config.hidden_size
        }
    }
    torch.save(model_dict, classifier_path)
    print(f"已保存分類器權重至 {classifier_path}")

def load_model(self, model_dir, device=None):
    """ 載入分類器權重 """
    if device is None:
        device = torch.device('cuda' if torch.cuda.is_available()
    else 'cpu')

    classifier_path = os.path.join(model_dir,
    "classifier_weights.pt")
    if os.path.exists(classifier_path):
        model_dict = torch.load(classifier_path,
        map_location=device, weights_only=True)

        # 載入各組件
        self.classifier.load_state_dict(model_dict['classifier'])
    self.attention_pooler.load_state_dict(model_dict['attention_pooler'])
    self.layer_weights.data =
    model_dict['layer_weights'].to(device)

    print(f"已載入分類器權重: {classifier_path}")
    return True
else:

```



```
print(f"警告：找不到分類器權重檔案 {classifier_path}")
return False
```

初始化模型

分類任務：兩者都可用，但 AutoModel 更輕量
生成功能：只有 AutoModelForCausalLM 支持
内存使用：AutoModelForCausalLM 通常較大，因為包含了完整的語言模型頭
在 Qwen2 情感分類模型中，使用 AutoModelForCausalLM 更為靈活，因為它既可以進行分類，也保留了原始的文本生成功能。

```
# AutoModelForCausalLM 輸出
outputs = base_model(input_ids, attention_mask)
# 輸出包含：
# - loss: (可選) 語言模型損失
# - logits: 張量，形狀為 [batch_size, sequence_length, vocab_size]
# - past_key_values: (可選) 用於加速解碼的過去狀態
# - hidden_states: (可選) 所有隱藏層狀態的元組
# - attentions: (可選) 注意力權重的元組

# AutoModel 輸出
outputs = base_model(input_ids, attention_mask)
# 輸出包含：
# - last_hidden_state: 張量，形狀為 [batch_size, sequence_length,
hidden_size]
# - hidden_states: (可選) 所有隱藏層狀態的元組
# - attentions: (可選) 注意力權重的元組

len(categories)

2

# 在外部先載入 base_model 預訓練權重(不包含分類層)
full_model = AutoModelForCausalLM.from_pretrained(model_id)
#
hidden_size = full_model.config.hidden_size
model = QwenForClassifier(full_model.model, hidden_size, num_labels=
len(categories))

# 移動到指定設備
model = model.to(device)

{"model_id": "f69cfb4079f344cb8a5fa63c6718dc28", "version_major": 2, "vers
ion_minor": 0}
```

Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better

performance, install the package with: ``pip install huggingface_hub[hf_xet]`` or ``pip install hf_xet``
WARNING:huggingface_hub.file_download:Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better performance, install the package with: ``pip install huggingface_hub[hf_xet]`` or ``pip install hf_xet``

```
{"model_id":"491a3e1397aa41b69d61d4488f529722","version_major":2,"version_minor":0}
```

Sliding Window Attention is enabled but not implemented for ``sdpa``; unexpected results may be encountered.

```
{"model_id":"c9e4074a932a463f8c6cd632fc428e6e","version_major":2,"version_minor":0}
```

full_model.model

```
Qwen2Model(
  (embed_tokens): Embedding(151936, 896)
  (layers): ModuleList(
    (0-23): 24 x Qwen2DecoderLayer(
      (self_attn): Qwen2Attention(
        (q_proj): Linear(in_features=896, out_features=896, bias=True)
        (k_proj): Linear(in_features=896, out_features=128, bias=True)
        (v_proj): Linear(in_features=896, out_features=128, bias=True)
        (o_proj): Linear(in_features=896, out_features=896,
bias=False)
      )
      (mlp): Qwen2MLP(
        (gate_proj): Linear(in_features=896, out_features=4864,
bias=False)
        (up_proj): Linear(in_features=896, out_features=4864,
bias=False)
        (down_proj): Linear(in_features=4864, out_features=896,
bias=False)
        (act_fn): SiLU()
      )
      (input_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
      (post_attention_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
    )
  )
  (norm): Qwen2RMSNorm((896,), eps=1e-06)
  (rotary_emb): Qwen2RotaryEmbedding()
)
```

full_model.base_model

```
Qwen2Model(
  (embed_tokens): Embedding(151936, 896)
  (layers): ModuleList(
```

```

        (0-23): 24 x Qwen2DecoderLayer(
          (self_attn): Qwen2Attention(
            (q_proj): Linear(in_features=896, out_features=896, bias=True)
            (k_proj): Linear(in_features=896, out_features=128, bias=True)
            (v_proj): Linear(in_features=896, out_features=128, bias=True)
            (o_proj): Linear(in_features=896, out_features=896,
bias=False)
          )
          (mlp): Qwen2MLP(
            (gate_proj): Linear(in_features=896, out_features=4864,
bias=False)
            (up_proj): Linear(in_features=896, out_features=4864,
bias=False)
            (down_proj): Linear(in_features=4864, out_features=896,
bias=False)
            (act_fn): SiLU()
          )
          (input_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
          (post_attention_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
        )
      )
      (norm): Qwen2RMSNorm((896,), eps=1e-06)
      (rotary_emb): Qwen2RotaryEmbedding()
    )

```

看看 base_model 與 model 有何不同?

model 內部有: 一個 base_model+輸出分類層。它被拼接為分類器，可以做兩個類別的分類

base_model 仍舊是 GPT 自回歸模型

但是在記憶體，兩者是共享同一份 base_model 權重，model 有多一個分類器的輸出層的部分

```

full_model
Qwen2ForCausalLM(
  (model): Qwen2Model(
    (embed_tokens): Embedding(151936, 896)
    (layers): ModuleList(
      (0-23): 24 x Qwen2DecoderLayer(
        (self_attn): Qwen2Attention(
          (q_proj): Linear(in_features=896, out_features=896,
bias=True)
          (k_proj): Linear(in_features=896, out_features=128,
bias=True)
          (v_proj): Linear(in_features=896, out_features=128,
bias=True)
          (o_proj): Linear(in_features=896, out_features=896,
bias=False)
        )
      )
    )
  )

```

```

        )
        (mlp): Qwen2MLP(
          (gate_proj): Linear(in_features=896, out_features=4864,
bias=False)
          (up_proj): Linear(in_features=896, out_features=4864,
bias=False)
          (down_proj): Linear(in_features=4864, out_features=896,
bias=False)
          (act_fn): SiLU()
        )
        (input_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
        (post_attention_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
      )
    )
    (norm): Qwen2RMSNorm((896,), eps=1e-06)
    (rotary_emb): Qwen2RotaryEmbedding()
  )
  (lm_head): Linear(in_features=896, out_features=151936, bias=False)
)

model

QwenForClassifier(
  (base_model): Qwen2Model(
    (embed_tokens): Embedding(151936, 896)
    (layers): ModuleList(
      (0-23): 24 x Qwen2DecoderLayer(
        (self_attn): Qwen2Attention(
          (q_proj): Linear(in_features=896, out_features=896,
bias=True)
          (k_proj): Linear(in_features=896, out_features=128,
bias=True)
          (v_proj): Linear(in_features=896, out_features=128,
bias=True)
          (o_proj): Linear(in_features=896, out_features=896,
bias=False)
        )
        (mlp): Qwen2MLP(
          (gate_proj): Linear(in_features=896, out_features=4864,
bias=False)
          (up_proj): Linear(in_features=896, out_features=4864,
bias=False)
          (down_proj): Linear(in_features=4864, out_features=896,
bias=False)
          (act_fn): SiLU()
        )
        (input_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
        (post_attention_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
      )
    )
  )
)

```

```

        (norm): Qwen2RMSNorm((896,), eps=1e-06)
        (rotary_emb): Qwen2RotaryEmbedding()
    )
    (attention_pooler): Sequential(
      (0): Linear(in_features=896, out_features=128, bias=True)
      (1): Tanh()
      (2): Linear(in_features=128, out_features=1, bias=True)
    )
    (classifier): Sequential(
      (0): Linear(in_features=896, out_features=256, bias=True)
      (1): LayerNorm((256,), eps=1e-05, elementwise_affine=True)
      (2): GELU(approximate='none')
      (3): Dropout(p=0.2, inplace=False)
      (4): Linear(in_features=256, out_features=128, bias=True)
      (5): LayerNorm((128,), eps=1e-05, elementwise_affine=True)
      (6): GELU(approximate='none')
      (7): Dropout(p=0.1, inplace=False)
      (8): Linear(in_features=128, out_features=2, bias=True)
    )
  )
)

# 這與model 是一樣的模型??
model.base_model

Qwen2Model(
  (embed_tokens): Embedding(151936, 896)
  (layers): ModuleList(
    (0-23): 24 x Qwen2DecoderLayer(
      (self_attn): Qwen2Attention(
        (q_proj): Linear(in_features=896, out_features=896, bias=True)
        (k_proj): Linear(in_features=896, out_features=128, bias=True)
        (v_proj): Linear(in_features=896, out_features=128, bias=True)
        (o_proj): Linear(in_features=896, out_features=896,
bias=False)
      )
      (mlp): Qwen2MLP(
        (gate_proj): Linear(in_features=896, out_features=4864,
bias=False)
        (up_proj): Linear(in_features=896, out_features=4864,
bias=False)
        (down_proj): Linear(in_features=4864, out_features=896,
bias=False)
        (act_fn): SiLU()
      )
      (input_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
      (post_attention_layernorm): Qwen2RMSNorm((896,), eps=1e-06)
    )
  )
  (norm): Qwen2RMSNorm((896,), eps=1e-06)
)

```

```
(rotary_emb): Qwen2RotaryEmbedding()  
)
```

模型怎麼用?

尚未訓練的模型 (classifier 的數據都是 0)

```
# Function to make predictions  
def predict_sentiment(text, model, tokenizer, device):  
    # Tokenize the input text  
    inputs = tokenizer(  
        text,  
        max_length=512,  
        truncation=True,  
        return_tensors="pt"  
    ).to(device)  
  
    # Get model predictions  
    with torch.no_grad():  
        outputs = model(**inputs)  
  
    # Extract logits and apply softmax to get probabilities  
    # logits = outputs.logits  
    logits = outputs["logits"] # 取出 logits  
  
    probabilities = torch.nn.functional.softmax(logits, dim=-1)  
  
    # Get the predicted class (0: negative, 1: positive)  
    predicted_class = torch.argmax(probabilities, dim=-1).item()  
  
    # Get the class name using id_to_label  
    predicted_label = id_to_label[predicted_class]  
  
    # Get the confidence score  
    confidence = probabilities[0][predicted_class].item()  
  
    return {  
        "text": text,  
        "sentiment": predicted_label,  
        "confidence": round(confidence, 2),  
        "probabilities": {  
            id_to_label[i]: round(prob.item(), 2) for i, prob in  
enumerate(probabilities[0])  
        }  
    }
```

```

text = "今天天氣真好，我很開心"
predict_sentiment(text, model, tokenizer, device)

{'text': '今天天氣真好，我很開心',
 'sentiment': '負面',
 'confidence': 0.53,
 'probabilities': {'負面': 0.53, '正面': 0.47}}

text = "這個產品品質差，服務更糟糕"
predict_sentiment(text, model, tokenizer, device)

{'text': '這個產品品質差，服務更糟糕',
 'sentiment': '負面',
 'confidence': 0.6,
 'probabilities': {'負面': 0.6, '正面': 0.4}}

text = "這家餐廳的食物美味，環境也很舒適"
predict_sentiment(text, model, tokenizer, device)

{'text': '這家餐廳的食物美味，環境也很舒適',
 'sentiment': '負面',
 'confidence': 0.74,
 'probabilities': {'負面': 0.74, '正面': 0.26}}

```

模型每個參數層都是 Trainable

```

def print_model_details(model):
    print("\n" + "="*80)
    print("MODEL ARCHITECTURE WITH TRAINABILITY STATUS")
    print("="*80)

    trainable_params = 0
    non_trainable_params = 0

    # Print each layer with details
    for idx, (name, layer) in enumerate(model.named_modules(), 1):
        if not list(layer.named_children()): # Only print leaf nodes
            (actual layers)
            param_count = sum(p.numel() for p in layer.parameters())
            status = "TRAINABLE" if any(p.requires_grad for p in
layer.parameters()) else "NON-TRAINABLE"

            print(f"\nLayer #{idx:02d}")
            print(f"├ Name: {name}")
            print(f"├ Type: {layer.__class__.__name__}")
            print(f"├ Details: {layer}")
            print(f"├ Parameters: {param_count:,}")
            print(f"└ Status: {status}")

            if status == "TRAINABLE":

```

```

        trainable_params += param_count
    else:
        non_trainable_params += param_count

    total_params = trainable_params + non_trainable_params
    trainable_percentage = (trainable_params / total_params) * 100 if
total_params > 0 else 0

    print("\n" + "="*80)
    print(f"Total Trainable Parameters: {trainable_params:,}")
    print(f"Total Non-Trainable Parameters: {non_trainable_params:,}")
    print(f"Total Parameters: {total_params:,}")
    print(f"Trainable Parameters Percentage:
{trainable_percentage:.2f}%")
    print("="*80)

print_model_details(model)

```

```

=====
=====
MODEL ARCHITECTURE WITH TRAINABILITY STATUS
=====
=====

```

Layer #03

```

├ Name: base_model.embed_tokens
├ Type: Embedding
├ Details: Embedding(151936, 896)
├ Parameters: 136,134,656
└ Status: NON-TRAINABLE

```

Layer #07

```

├ Name: base_model.layers.0.self_attn.q_proj
├ Type: Linear
├ Details: Linear(in_features=896, out_features=896, bias=True)
├ Parameters: 803,712
└ Status: NON-TRAINABLE

```

Layer #08

```

├ Name: base_model.layers.0.self_attn.k_proj
├ Type: Linear
├ Details: Linear(in_features=896, out_features=128, bias=True)
├ Parameters: 114,816
└ Status: NON-TRAINABLE

```

Layer #09

```

├ Name: base_model.layers.0.self_attn.v_proj
├ Type: Linear
├ Details: Linear(in_features=896, out_features=128, bias=True)

```


- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #10

- └ Name: base_model.layers.0.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #12

- └ Name: base_model.layers.0.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #13

- └ Name: base_model.layers.0.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #14

- └ Name: base_model.layers.0.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #15

- └ Name: base_model.layers.0.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #16

- └ Name: base_model.layers.0.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #17

- └ Name: base_model.layers.0.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896

└─ Status: NON-TRAINABLE

Layer #20

└─ Name: base_model.layers.1.self_attn.q_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=True)
└─ Parameters: 803,712
└─ Status: NON-TRAINABLE

Layer #21

└─ Name: base_model.layers.1.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #22

└─ Name: base_model.layers.1.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #23

└─ Name: base_model.layers.1.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #25

└─ Name: base_model.layers.1.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #26

└─ Name: base_model.layers.1.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #27

└─ Name: base_model.layers.1.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #28

- └ Name: base_model.layers.1.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #29

- └ Name: base_model.layers.1.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #30

- └ Name: base_model.layers.1.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #33

- └ Name: base_model.layers.2.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #34

- └ Name: base_model.layers.2.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #35

- └ Name: base_model.layers.2.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #36

- └ Name: base_model.layers.2.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #38

- └ Name: base_model.layers.2.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #39

- └ Name: base_model.layers.2.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #40

- └ Name: base_model.layers.2.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #41

- └ Name: base_model.layers.2.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #42

- └ Name: base_model.layers.2.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #43

- └ Name: base_model.layers.2.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #46

- └ Name: base_model.layers.3.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #47

- └ Name: base_model.layers.3.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #48

- └ Name: base_model.layers.3.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #49

- └ Name: base_model.layers.3.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #51

- └ Name: base_model.layers.3.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #52

- └ Name: base_model.layers.3.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #53

- └ Name: base_model.layers.3.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #54

- └ Name: base_model.layers.3.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #55

- └ Name: base_model.layers.3.input_layernorm

- | Type: Qwen2RMSNorm
- | Details: Qwen2RMSNorm((896,), eps=1e-06)
- | Parameters: 896
- | Status: NON-TRAINABLE

Layer #56

- | Name: base_model.layers.3.post_attention_layernorm
- | Type: Qwen2RMSNorm
- | Details: Qwen2RMSNorm((896,), eps=1e-06)
- | Parameters: 896
- | Status: NON-TRAINABLE

Layer #59

- | Name: base_model.layers.4.self_attn.q_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=896, bias=True)
- | Parameters: 803,712
- | Status: NON-TRAINABLE

Layer #60

- | Name: base_model.layers.4.self_attn.k_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: NON-TRAINABLE

Layer #61

- | Name: base_model.layers.4.self_attn.v_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: NON-TRAINABLE

Layer #62

- | Name: base_model.layers.4.self_attn.o_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=896, bias=False)
- | Parameters: 802,816
- | Status: NON-TRAINABLE

Layer #64

- | Name: base_model.layers.4.mlp.gate_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #65

- | Name: base_model.layers.4.mlp.up_proj
- | Type: Linear

- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #66

- └ Name: base_model.layers.4.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #67

- └ Name: base_model.layers.4.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #68

- └ Name: base_model.layers.4.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #69

- └ Name: base_model.layers.4.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #72

- └ Name: base_model.layers.5.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #73

- └ Name: base_model.layers.5.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #74

- └ Name: base_model.layers.5.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)

- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #75

- └ Name: base_model.layers.5.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #77

- └ Name: base_model.layers.5.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #78

- └ Name: base_model.layers.5.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #79

- └ Name: base_model.layers.5.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #80

- └ Name: base_model.layers.5.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #81

- └ Name: base_model.layers.5.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #82

- └ Name: base_model.layers.5.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896

└─ Status: NON-TRAINABLE

Layer #85

└─ Name: base_model.layers.6.self_attn.q_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=True)
└─ Parameters: 803,712
└─ Status: NON-TRAINABLE

Layer #86

└─ Name: base_model.layers.6.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #87

└─ Name: base_model.layers.6.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #88

└─ Name: base_model.layers.6.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #90

└─ Name: base_model.layers.6.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #91

└─ Name: base_model.layers.6.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #92

└─ Name: base_model.layers.6.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #93

- └ Name: base_model.layers.6.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #94

- └ Name: base_model.layers.6.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #95

- └ Name: base_model.layers.6.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #98

- └ Name: base_model.layers.7.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #99

- └ Name: base_model.layers.7.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #100

- └ Name: base_model.layers.7.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #101

- └ Name: base_model.layers.7.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #103

- └ Name: base_model.layers.7.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #104

- └ Name: base_model.layers.7.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #105

- └ Name: base_model.layers.7.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #106

- └ Name: base_model.layers.7.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #107

- └ Name: base_model.layers.7.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #108

- └ Name: base_model.layers.7.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #111

- └ Name: base_model.layers.8.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #112

- └ Name: base_model.layers.8.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #113

- └ Name: base_model.layers.8.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #114

- └ Name: base_model.layers.8.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #116

- └ Name: base_model.layers.8.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #117

- └ Name: base_model.layers.8.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #118

- └ Name: base_model.layers.8.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #119

- └ Name: base_model.layers.8.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #120

- └ Name: base_model.layers.8.input_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #121

- └ Name: base_model.layers.8.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #124

- └ Name: base_model.layers.9.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #125

- └ Name: base_model.layers.9.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #126

- └ Name: base_model.layers.9.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #127

- └ Name: base_model.layers.9.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #129

- └ Name: base_model.layers.9.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #130

- └ Name: base_model.layers.9.mlp.up_proj
- └ Type: Linear

- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #131

- └ Name: base_model.layers.9.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #132

- └ Name: base_model.layers.9.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #133

- └ Name: base_model.layers.9.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #134

- └ Name: base_model.layers.9.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #137

- └ Name: base_model.layers.10.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #138

- └ Name: base_model.layers.10.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #139

- └ Name: base_model.layers.10.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)

- Parameters: 114,816
- Status: NON-TRAINABLE

Layer #140

- Name: `base_model.layers.10.self_attn.o_proj`
- Type: Linear
- Details: `Linear(in_features=896, out_features=896, bias=False)`
- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #142

- Name: `base_model.layers.10.mlp.gate_proj`
- Type: Linear
- Details: `Linear(in_features=896, out_features=4864, bias=False)`
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #143

- Name: `base_model.layers.10.mlp.up_proj`
- Type: Linear
- Details: `Linear(in_features=896, out_features=4864, bias=False)`
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #144

- Name: `base_model.layers.10.mlp.down_proj`
- Type: Linear
- Details: `Linear(in_features=4864, out_features=896, bias=False)`
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #145

- Name: `base_model.layers.10.mlp.act_fn`
- Type: SiLU
- Details: `SiLU()`
- Parameters: 0
- Status: NON-TRAINABLE

Layer #146

- Name: `base_model.layers.10.input_layernorm`
- Type: Qwen2RMSNorm
- Details: `Qwen2RMSNorm((896,), eps=1e-06)`
- Parameters: 896
- Status: NON-TRAINABLE

Layer #147

- Name: `base_model.layers.10.post_attention_layernorm`
- Type: Qwen2RMSNorm
- Details: `Qwen2RMSNorm((896,), eps=1e-06)`
- Parameters: 896

└─ Status: NON-TRAINABLE

Layer #150

└─ Name: base_model.layers.11.self_attn.q_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=True)
└─ Parameters: 803,712
└─ Status: NON-TRAINABLE

Layer #151

└─ Name: base_model.layers.11.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #152

└─ Name: base_model.layers.11.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #153

└─ Name: base_model.layers.11.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #155

└─ Name: base_model.layers.11.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #156

└─ Name: base_model.layers.11.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #157

└─ Name: base_model.layers.11.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #158

- └ Name: base_model.layers.11.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #159

- └ Name: base_model.layers.11.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #160

- └ Name: base_model.layers.11.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #163

- └ Name: base_model.layers.12.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #164

- └ Name: base_model.layers.12.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #165

- └ Name: base_model.layers.12.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #166

- └ Name: base_model.layers.12.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #168

- └ Name: base_model.layers.12.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #169

- └ Name: base_model.layers.12.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #170

- └ Name: base_model.layers.12.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #171

- └ Name: base_model.layers.12.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #172

- └ Name: base_model.layers.12.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #173

- └ Name: base_model.layers.12.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #176

- └ Name: base_model.layers.13.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #177

- └ Name: base_model.layers.13.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #178

- └ Name: base_model.layers.13.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #179

- └ Name: base_model.layers.13.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #181

- └ Name: base_model.layers.13.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #182

- └ Name: base_model.layers.13.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #183

- └ Name: base_model.layers.13.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #184

- └ Name: base_model.layers.13.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #185

- └ Name: base_model.layers.13.input_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #186

- └ Name: base_model.layers.13.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #189

- └ Name: base_model.layers.14.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #190

- └ Name: base_model.layers.14.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #191

- └ Name: base_model.layers.14.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #192

- └ Name: base_model.layers.14.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #194

- └ Name: base_model.layers.14.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #195

- └ Name: base_model.layers.14.mlp.up_proj
- └ Type: Linear

- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #196

- └ Name: base_model.layers.14.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #197

- └ Name: base_model.layers.14.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #198

- └ Name: base_model.layers.14.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #199

- └ Name: base_model.layers.14.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #202

- └ Name: base_model.layers.15.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #203

- └ Name: base_model.layers.15.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #204

- └ Name: base_model.layers.15.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)

- Parameters: 114,816
- Status: NON-TRAINABLE

Layer #205

- Name: base_model.layers.15.self_attn.o_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=896, bias=False)
- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #207

- Name: base_model.layers.15.mlp.gate_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #208

- Name: base_model.layers.15.mlp.up_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #209

- Name: base_model.layers.15.mlp.down_proj
- Type: Linear
- Details: Linear(in_features=4864, out_features=896, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #210

- Name: base_model.layers.15.mlp.act_fn
- Type: SiLU
- Details: SiLU()
- Parameters: 0
- Status: NON-TRAINABLE

Layer #211

- Name: base_model.layers.15.input_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,)), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #212

- Name: base_model.layers.15.post_attention_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,)), eps=1e-06)
- Parameters: 896

└─ Status: NON-TRAINABLE

Layer #215

└─ Name: base_model.layers.16.self_attn.q_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=True)
└─ Parameters: 803,712
└─ Status: NON-TRAINABLE

Layer #216

└─ Name: base_model.layers.16.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #217

└─ Name: base_model.layers.16.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #218

└─ Name: base_model.layers.16.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #220

└─ Name: base_model.layers.16.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #221

└─ Name: base_model.layers.16.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #222

└─ Name: base_model.layers.16.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #223

- └ Name: base_model.layers.16.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #224

- └ Name: base_model.layers.16.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #225

- └ Name: base_model.layers.16.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #228

- └ Name: base_model.layers.17.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #229

- └ Name: base_model.layers.17.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #230

- └ Name: base_model.layers.17.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #231

- └ Name: base_model.layers.17.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #233

- └ Name: base_model.layers.17.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #234

- └ Name: base_model.layers.17.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #235

- └ Name: base_model.layers.17.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #236

- └ Name: base_model.layers.17.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #237

- └ Name: base_model.layers.17.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #238

- └ Name: base_model.layers.17.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #241

- └ Name: base_model.layers.18.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #242

- | Name: base_model.layers.18.self_attn.k_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: NON-TRAINABLE

Layer #243

- | Name: base_model.layers.18.self_attn.v_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: NON-TRAINABLE

Layer #244

- | Name: base_model.layers.18.self_attn.o_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=896, bias=False)
- | Parameters: 802,816
- | Status: NON-TRAINABLE

Layer #246

- | Name: base_model.layers.18.mlp.gate_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #247

- | Name: base_model.layers.18.mlp.up_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #248

- | Name: base_model.layers.18.mlp.down_proj
- | Type: Linear
- | Details: Linear(in_features=4864, out_features=896, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #249

- | Name: base_model.layers.18.mlp.act_fn
- | Type: SiLU
- | Details: SiLU()
- | Parameters: 0
- | Status: NON-TRAINABLE

Layer #250

- | Name: base_model.layers.18.input_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #251

- └ Name: base_model.layers.18.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #254

- └ Name: base_model.layers.19.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #255

- └ Name: base_model.layers.19.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #256

- └ Name: base_model.layers.19.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #257

- └ Name: base_model.layers.19.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #259

- └ Name: base_model.layers.19.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #260

- └ Name: base_model.layers.19.mlp.up_proj
- └ Type: Linear

- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #261

- └ Name: base_model.layers.19.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #262

- └ Name: base_model.layers.19.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #263

- └ Name: base_model.layers.19.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #264

- └ Name: base_model.layers.19.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #267

- └ Name: base_model.layers.20.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #268

- └ Name: base_model.layers.20.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #269

- └ Name: base_model.layers.20.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)

- Parameters: 114,816
- Status: NON-TRAINABLE

Layer #270

- Name: base_model.layers.20.self_attn.o_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=896, bias=False)
- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #272

- Name: base_model.layers.20.mlp.gate_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #273

- Name: base_model.layers.20.mlp.up_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #274

- Name: base_model.layers.20.mlp.down_proj
- Type: Linear
- Details: Linear(in_features=4864, out_features=896, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #275

- Name: base_model.layers.20.mlp.act_fn
- Type: SiLU
- Details: SiLU()
- Parameters: 0
- Status: NON-TRAINABLE

Layer #276

- Name: base_model.layers.20.input_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,)), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #277

- Name: base_model.layers.20.post_attention_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,)), eps=1e-06)
- Parameters: 896

└─ Status: NON-TRAINABLE

Layer #280

└─ Name: base_model.layers.21.self_attn.q_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=True)
└─ Parameters: 803,712
└─ Status: NON-TRAINABLE

Layer #281

└─ Name: base_model.layers.21.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #282

└─ Name: base_model.layers.21.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #283

└─ Name: base_model.layers.21.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #285

└─ Name: base_model.layers.21.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #286

└─ Name: base_model.layers.21.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #287

└─ Name: base_model.layers.21.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #288

- └ Name: base_model.layers.21.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #289

- └ Name: base_model.layers.21.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #290

- └ Name: base_model.layers.21.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #293

- └ Name: base_model.layers.22.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #294

- └ Name: base_model.layers.22.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #295

- └ Name: base_model.layers.22.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #296

- └ Name: base_model.layers.22.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #298

- └ Name: `base_model.layers.22.mlp.gate_proj`
- └ Type: Linear
- └ Details: `Linear(in_features=896, out_features=4864, bias=False)`
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #299

- └ Name: `base_model.layers.22.mlp.up_proj`
- └ Type: Linear
- └ Details: `Linear(in_features=896, out_features=4864, bias=False)`
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #300

- └ Name: `base_model.layers.22.mlp.down_proj`
- └ Type: Linear
- └ Details: `Linear(in_features=4864, out_features=896, bias=False)`
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #301

- └ Name: `base_model.layers.22.mlp.act_fn`
- └ Type: SiLU
- └ Details: `SiLU()`
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #302

- └ Name: `base_model.layers.22.input_layernorm`
- └ Type: Qwen2RMSNorm
- └ Details: `Qwen2RMSNorm((896,), eps=1e-06)`
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #303

- └ Name: `base_model.layers.22.post_attention_layernorm`
- └ Type: Qwen2RMSNorm
- └ Details: `Qwen2RMSNorm((896,), eps=1e-06)`
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #306

- └ Name: `base_model.layers.23.self_attn.q_proj`
- └ Type: Linear
- └ Details: `Linear(in_features=896, out_features=896, bias=True)`
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #307

- └ Name: base_model.layers.23.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #308

- └ Name: base_model.layers.23.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #309

- └ Name: base_model.layers.23.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #311

- └ Name: base_model.layers.23.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #312

- └ Name: base_model.layers.23.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #313

- └ Name: base_model.layers.23.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #314

- └ Name: base_model.layers.23.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #315

- └ Name: base_model.layers.23.input_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #316

- └ Name: base_model.layers.23.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #317

- └ Name: base_model.norm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #318

- └ Name: base_model.rotary_emb
- └ Type: Qwen2RotaryEmbedding
- └ Details: Qwen2RotaryEmbedding()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #320

- └ Name: attention_pooler.0
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: TRAINABLE

Layer #321

- └ Name: attention_pooler.1
- └ Type: Tanh
- └ Details: Tanh()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #322

- └ Name: attention_pooler.2
- └ Type: Linear
- └ Details: Linear(in_features=128, out_features=1, bias=True)
- └ Parameters: 129
- └ Status: TRAINABLE

Layer #324

- └ Name: classifier.0
- └ Type: Linear

- └ Details: Linear(in_features=896, out_features=256, bias=True)
- └ Parameters: 229,632
- └ Status: TRAINABLE

Layer #325

- └ Name: classifier.1
- └ Type: LayerNorm
- └ Details: LayerNorm((256,), eps=1e-05, elementwise_affine=True)
- └ Parameters: 512
- └ Status: TRAINABLE

Layer #326

- └ Name: classifier.2
- └ Type: GELU
- └ Details: GELU(approximate='none')
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #327

- └ Name: classifier.3
- └ Type: Dropout
- └ Details: Dropout(p=0.2, inplace=False)
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #328

- └ Name: classifier.4
- └ Type: Linear
- └ Details: Linear(in_features=256, out_features=128, bias=True)
- └ Parameters: 32,896
- └ Status: TRAINABLE

Layer #329

- └ Name: classifier.5
- └ Type: LayerNorm
- └ Details: LayerNorm((128,), eps=1e-05, elementwise_affine=True)
- └ Parameters: 256
- └ Status: TRAINABLE

Layer #330

- └ Name: classifier.6
- └ Type: GELU
- └ Details: GELU(approximate='none')
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #331

- └ Name: classifier.7
- └ Type: Dropout
- └ Details: Dropout(p=0.1, inplace=False)

```
└ Parameters: 0
└ Status: NON-TRAINABLE
```

Layer #332

```
└ Name: classifier.8
└ Type: Linear
└ Details: Linear(in_features=128, out_features=2, bias=True)
└ Parameters: 258
└ Status: TRAINABLE
```

```
=====
=====
Total Trainable Parameters: 378,499
Total Non-Trainable Parameters: 494,032,768
Total Parameters: 494,411,267
Trainable Parameters Percentage: 0.08%
=====
=====
```

```
# 檢查原始參數是否可訓練
# print("Classifier weight requires_grad:",
# model.classifier.weight.requires_grad)
```

模型主體固定參數，只有極少數參數是 Trainable

```
#model.print_trainable_parameters()
```

```
def print_trainable_parameters(model):
    trainable_params = sum(p.numel() for p in model.parameters() if
p.requires_grad)
    all_params = sum(p.numel() for p in model.parameters())
    print(f"trainable params: {trainable_params:,d} || all params:
{all_params:,d} || trainable%: {100 * trainable_params /
all_params:.2f}%")
```

```
print_trainable_parameters(model)
```

```
trainable params: 378,503 || all params: 494,411,271 || trainable%:
0.08%
```

```
print_model_details(model)
```

```
=====
=====
MODEL ARCHITECTURE WITH TRAINABILITY STATUS
=====
=====
```

Layer #03

```
└ Name: base_model.embed_tokens
```

- └ Type: Embedding
- └ Details: Embedding(151936, 896)
- └ Parameters: 136,134,656
- └ Status: NON-TRAINABLE

Layer #07

- └ Name: base_model.layers.0.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #08

- └ Name: base_model.layers.0.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #09

- └ Name: base_model.layers.0.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #10

- └ Name: base_model.layers.0.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #12

- └ Name: base_model.layers.0.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #13

- └ Name: base_model.layers.0.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #14

- └ Name: base_model.layers.0.mlp.down_proj
- └ Type: Linear

- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #15

- └ Name: base_model.layers.0.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #16

- └ Name: base_model.layers.0.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #17

- └ Name: base_model.layers.0.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #20

- └ Name: base_model.layers.1.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #21

- └ Name: base_model.layers.1.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #22

- └ Name: base_model.layers.1.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #23

- └ Name: base_model.layers.1.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)

- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #25

- Name: base_model.layers.1.mlp.gate_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #26

- Name: base_model.layers.1.mlp.up_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #27

- Name: base_model.layers.1.mlp.down_proj
- Type: Linear
- Details: Linear(in_features=4864, out_features=896, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #28

- Name: base_model.layers.1.mlp.act_fn
- Type: SiLU
- Details: SiLU()
- Parameters: 0
- Status: NON-TRAINABLE

Layer #29

- Name: base_model.layers.1.input_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #30

- Name: base_model.layers.1.post_attention_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #33

- Name: base_model.layers.2.self_attn.q_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=896, bias=True)
- Parameters: 803,712

└─ Status: NON-TRAINABLE

Layer #34

└─ Name: base_model.layers.2.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #35

└─ Name: base_model.layers.2.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #36

└─ Name: base_model.layers.2.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #38

└─ Name: base_model.layers.2.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #39

└─ Name: base_model.layers.2.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #40

└─ Name: base_model.layers.2.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #41

└─ Name: base_model.layers.2.mlp.act_fn
└─ Type: SiLU
└─ Details: SiLU()
└─ Parameters: 0
└─ Status: NON-TRAINABLE

Layer #42

- └ Name: base_model.layers.2.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #43

- └ Name: base_model.layers.2.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #46

- └ Name: base_model.layers.3.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #47

- └ Name: base_model.layers.3.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #48

- └ Name: base_model.layers.3.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #49

- └ Name: base_model.layers.3.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #51

- └ Name: base_model.layers.3.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #52

- └ Name: base_model.layers.3.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #53

- └ Name: base_model.layers.3.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #54

- └ Name: base_model.layers.3.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #55

- └ Name: base_model.layers.3.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #56

- └ Name: base_model.layers.3.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #59

- └ Name: base_model.layers.4.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #60

- └ Name: base_model.layers.4.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #61

- └ Name: base_model.layers.4.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #62

- └ Name: base_model.layers.4.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #64

- └ Name: base_model.layers.4.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #65

- └ Name: base_model.layers.4.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #66

- └ Name: base_model.layers.4.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #67

- └ Name: base_model.layers.4.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #68

- └ Name: base_model.layers.4.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #69

- └ Name: base_model.layers.4.post_attention_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #72

- └ Name: base_model.layers.5.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #73

- └ Name: base_model.layers.5.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #74

- └ Name: base_model.layers.5.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #75

- └ Name: base_model.layers.5.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #77

- └ Name: base_model.layers.5.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #78

- └ Name: base_model.layers.5.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #79

- └ Name: base_model.layers.5.mlp.down_proj
- └ Type: Linear

- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #80

- └ Name: base_model.layers.5.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #81

- └ Name: base_model.layers.5.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #82

- └ Name: base_model.layers.5.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #85

- └ Name: base_model.layers.6.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #86

- └ Name: base_model.layers.6.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #87

- └ Name: base_model.layers.6.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #88

- └ Name: base_model.layers.6.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)

- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #90

- Name: base_model.layers.6.mlp.gate_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #91

- Name: base_model.layers.6.mlp.up_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #92

- Name: base_model.layers.6.mlp.down_proj
- Type: Linear
- Details: Linear(in_features=4864, out_features=896, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #93

- Name: base_model.layers.6.mlp.act_fn
- Type: SiLU
- Details: SiLU()
- Parameters: 0
- Status: NON-TRAINABLE

Layer #94

- Name: base_model.layers.6.input_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #95

- Name: base_model.layers.6.post_attention_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #98

- Name: base_model.layers.7.self_attn.q_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=896, bias=True)
- Parameters: 803,712

└─ Status: NON-TRAINABLE

Layer #99

└─ Name: base_model.layers.7.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #100

└─ Name: base_model.layers.7.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #101

└─ Name: base_model.layers.7.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #103

└─ Name: base_model.layers.7.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #104

└─ Name: base_model.layers.7.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #105

└─ Name: base_model.layers.7.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #106

└─ Name: base_model.layers.7.mlp.act_fn
└─ Type: SiLU
└─ Details: SiLU()
└─ Parameters: 0
└─ Status: NON-TRAINABLE

Layer #107

- └ Name: base_model.layers.7.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #108

- └ Name: base_model.layers.7.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #111

- └ Name: base_model.layers.8.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #112

- └ Name: base_model.layers.8.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #113

- └ Name: base_model.layers.8.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #114

- └ Name: base_model.layers.8.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #116

- └ Name: base_model.layers.8.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #117

- └ Name: base_model.layers.8.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #118

- └ Name: base_model.layers.8.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #119

- └ Name: base_model.layers.8.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #120

- └ Name: base_model.layers.8.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #121

- └ Name: base_model.layers.8.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #124

- └ Name: base_model.layers.9.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #125

- └ Name: base_model.layers.9.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #126

- └ Name: base_model.layers.9.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #127

- └ Name: base_model.layers.9.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #129

- └ Name: base_model.layers.9.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #130

- └ Name: base_model.layers.9.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #131

- └ Name: base_model.layers.9.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #132

- └ Name: base_model.layers.9.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #133

- └ Name: base_model.layers.9.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #134

- └ Name: base_model.layers.9.post_attention_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #137

- └ Name: base_model.layers.10.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #138

- └ Name: base_model.layers.10.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #139

- └ Name: base_model.layers.10.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #140

- └ Name: base_model.layers.10.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #142

- └ Name: base_model.layers.10.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #143

- └ Name: base_model.layers.10.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #144

- └ Name: base_model.layers.10.mlp.down_proj
- └ Type: Linear

- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #145

- └ Name: base_model.layers.10.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #146

- └ Name: base_model.layers.10.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #147

- └ Name: base_model.layers.10.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #150

- └ Name: base_model.layers.11.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #151

- └ Name: base_model.layers.11.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #152

- └ Name: base_model.layers.11.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #153

- └ Name: base_model.layers.11.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)

- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #155

- Name: base_model.layers.11.mlp.gate_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #156

- Name: base_model.layers.11.mlp.up_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #157

- Name: base_model.layers.11.mlp.down_proj
- Type: Linear
- Details: Linear(in_features=4864, out_features=896, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #158

- Name: base_model.layers.11.mlp.act_fn
- Type: SiLU
- Details: SiLU()
- Parameters: 0
- Status: NON-TRAINABLE

Layer #159

- Name: base_model.layers.11.input_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #160

- Name: base_model.layers.11.post_attention_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #163

- Name: base_model.layers.12.self_attn.q_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=896, bias=True)
- Parameters: 803,712

└─ Status: NON-TRAINABLE

Layer #164

└─ Name: base_model.layers.12.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #165

└─ Name: base_model.layers.12.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #166

└─ Name: base_model.layers.12.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #168

└─ Name: base_model.layers.12.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #169

└─ Name: base_model.layers.12.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #170

└─ Name: base_model.layers.12.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #171

└─ Name: base_model.layers.12.mlp.act_fn
└─ Type: SiLU
└─ Details: SiLU()
└─ Parameters: 0
└─ Status: NON-TRAINABLE

Layer #172

- └ Name: base_model.layers.12.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #173

- └ Name: base_model.layers.12.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #176

- └ Name: base_model.layers.13.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #177

- └ Name: base_model.layers.13.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #178

- └ Name: base_model.layers.13.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #179

- └ Name: base_model.layers.13.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #181

- └ Name: base_model.layers.13.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #182

- └ Name: base_model.layers.13.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #183

- └ Name: base_model.layers.13.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #184

- └ Name: base_model.layers.13.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #185

- └ Name: base_model.layers.13.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #186

- └ Name: base_model.layers.13.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #189

- └ Name: base_model.layers.14.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #190

- └ Name: base_model.layers.14.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #191

- | Name: base_model.layers.14.self_attn.v_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: NON-TRAINABLE

Layer #192

- | Name: base_model.layers.14.self_attn.o_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=896, bias=False)
- | Parameters: 802,816
- | Status: NON-TRAINABLE

Layer #194

- | Name: base_model.layers.14.mlp.gate_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #195

- | Name: base_model.layers.14.mlp.up_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #196

- | Name: base_model.layers.14.mlp.down_proj
- | Type: Linear
- | Details: Linear(in_features=4864, out_features=896, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #197

- | Name: base_model.layers.14.mlp.act_fn
- | Type: SiLU
- | Details: SiLU()
- | Parameters: 0
- | Status: NON-TRAINABLE

Layer #198

- | Name: base_model.layers.14.input_layernorm
- | Type: Qwen2RMSNorm
- | Details: Qwen2RMSNorm((896,), eps=1e-06)
- | Parameters: 896
- | Status: NON-TRAINABLE

Layer #199

- | Name: base_model.layers.14.post_attention_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #202

- └ Name: base_model.layers.15.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #203

- └ Name: base_model.layers.15.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #204

- └ Name: base_model.layers.15.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #205

- └ Name: base_model.layers.15.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #207

- └ Name: base_model.layers.15.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #208

- └ Name: base_model.layers.15.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #209

- └ Name: base_model.layers.15.mlp.down_proj
- └ Type: Linear

- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #210

- └ Name: base_model.layers.15.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #211

- └ Name: base_model.layers.15.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #212

- └ Name: base_model.layers.15.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #215

- └ Name: base_model.layers.16.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #216

- └ Name: base_model.layers.16.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #217

- └ Name: base_model.layers.16.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #218

- └ Name: base_model.layers.16.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)

- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #220

- Name: base_model.layers.16.mlp.gate_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #221

- Name: base_model.layers.16.mlp.up_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=4864, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #222

- Name: base_model.layers.16.mlp.down_proj
- Type: Linear
- Details: Linear(in_features=4864, out_features=896, bias=False)
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #223

- Name: base_model.layers.16.mlp.act_fn
- Type: SiLU
- Details: SiLU()
- Parameters: 0
- Status: NON-TRAINABLE

Layer #224

- Name: base_model.layers.16.input_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #225

- Name: base_model.layers.16.post_attention_layernorm
- Type: Qwen2RMSNorm
- Details: Qwen2RMSNorm((896,), eps=1e-06)
- Parameters: 896
- Status: NON-TRAINABLE

Layer #228

- Name: base_model.layers.17.self_attn.q_proj
- Type: Linear
- Details: Linear(in_features=896, out_features=896, bias=True)
- Parameters: 803,712

└─ Status: NON-TRAINABLE

Layer #229

└─ Name: base_model.layers.17.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #230

└─ Name: base_model.layers.17.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #231

└─ Name: base_model.layers.17.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #233

└─ Name: base_model.layers.17.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #234

└─ Name: base_model.layers.17.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #235

└─ Name: base_model.layers.17.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #236

└─ Name: base_model.layers.17.mlp.act_fn
└─ Type: SiLU
└─ Details: SiLU()
└─ Parameters: 0
└─ Status: NON-TRAINABLE

Layer #237

- └ Name: base_model.layers.17.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #238

- └ Name: base_model.layers.17.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #241

- └ Name: base_model.layers.18.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #242

- └ Name: base_model.layers.18.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #243

- └ Name: base_model.layers.18.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #244

- └ Name: base_model.layers.18.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #246

- └ Name: base_model.layers.18.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #247

- └ Name: base_model.layers.18.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #248

- └ Name: base_model.layers.18.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #249

- └ Name: base_model.layers.18.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #250

- └ Name: base_model.layers.18.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #251

- └ Name: base_model.layers.18.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #254

- └ Name: base_model.layers.19.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #255

- └ Name: base_model.layers.19.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #256

- | Name: base_model.layers.19.self_attn.v_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: NON-TRAINABLE

Layer #257

- | Name: base_model.layers.19.self_attn.o_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=896, bias=False)
- | Parameters: 802,816
- | Status: NON-TRAINABLE

Layer #259

- | Name: base_model.layers.19.mlp.gate_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #260

- | Name: base_model.layers.19.mlp.up_proj
- | Type: Linear
- | Details: Linear(in_features=896, out_features=4864, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #261

- | Name: base_model.layers.19.mlp.down_proj
- | Type: Linear
- | Details: Linear(in_features=4864, out_features=896, bias=False)
- | Parameters: 4,358,144
- | Status: NON-TRAINABLE

Layer #262

- | Name: base_model.layers.19.mlp.act_fn
- | Type: SiLU
- | Details: SiLU()
- | Parameters: 0
- | Status: NON-TRAINABLE

Layer #263

- | Name: base_model.layers.19.input_layernorm
- | Type: Qwen2RMSNorm
- | Details: Qwen2RMSNorm((896,), eps=1e-06)
- | Parameters: 896
- | Status: NON-TRAINABLE

Layer #264

- | Name: base_model.layers.19.post_attention_layernorm

- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #267

- └ Name: base_model.layers.20.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #268

- └ Name: base_model.layers.20.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #269

- └ Name: base_model.layers.20.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #270

- └ Name: base_model.layers.20.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #272

- └ Name: base_model.layers.20.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #273

- └ Name: base_model.layers.20.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #274

- └ Name: base_model.layers.20.mlp.down_proj
- └ Type: Linear

- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #275

- └ Name: base_model.layers.20.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #276

- └ Name: base_model.layers.20.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #277

- └ Name: base_model.layers.20.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,)), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #280

- └ Name: base_model.layers.21.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #281

- └ Name: base_model.layers.21.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #282

- └ Name: base_model.layers.21.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #283

- └ Name: base_model.layers.21.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)

- Parameters: 802,816
- Status: NON-TRAINABLE

Layer #285

- Name: `base_model.layers.21.mlp.gate_proj`
- Type: Linear
- Details: `Linear(in_features=896, out_features=4864, bias=False)`
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #286

- Name: `base_model.layers.21.mlp.up_proj`
- Type: Linear
- Details: `Linear(in_features=896, out_features=4864, bias=False)`
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #287

- Name: `base_model.layers.21.mlp.down_proj`
- Type: Linear
- Details: `Linear(in_features=4864, out_features=896, bias=False)`
- Parameters: 4,358,144
- Status: NON-TRAINABLE

Layer #288

- Name: `base_model.layers.21.mlp.act_fn`
- Type: SiLU
- Details: `SiLU()`
- Parameters: 0
- Status: NON-TRAINABLE

Layer #289

- Name: `base_model.layers.21.input_layernorm`
- Type: `Qwen2RMSNorm`
- Details: `Qwen2RMSNorm((896,), eps=1e-06)`
- Parameters: 896
- Status: NON-TRAINABLE

Layer #290

- Name: `base_model.layers.21.post_attention_layernorm`
- Type: `Qwen2RMSNorm`
- Details: `Qwen2RMSNorm((896,), eps=1e-06)`
- Parameters: 896
- Status: NON-TRAINABLE

Layer #293

- Name: `base_model.layers.22.self_attn.q_proj`
- Type: Linear
- Details: `Linear(in_features=896, out_features=896, bias=True)`
- Parameters: 803,712

└─ Status: NON-TRAINABLE

Layer #294

└─ Name: base_model.layers.22.self_attn.k_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #295

└─ Name: base_model.layers.22.self_attn.v_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=128, bias=True)
└─ Parameters: 114,816
└─ Status: NON-TRAINABLE

Layer #296

└─ Name: base_model.layers.22.self_attn.o_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=896, bias=False)
└─ Parameters: 802,816
└─ Status: NON-TRAINABLE

Layer #298

└─ Name: base_model.layers.22.mlp.gate_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #299

└─ Name: base_model.layers.22.mlp.up_proj
└─ Type: Linear
└─ Details: Linear(in_features=896, out_features=4864, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #300

└─ Name: base_model.layers.22.mlp.down_proj
└─ Type: Linear
└─ Details: Linear(in_features=4864, out_features=896, bias=False)
└─ Parameters: 4,358,144
└─ Status: NON-TRAINABLE

Layer #301

└─ Name: base_model.layers.22.mlp.act_fn
└─ Type: SiLU
└─ Details: SiLU()
└─ Parameters: 0
└─ Status: NON-TRAINABLE

Layer #302

- └ Name: base_model.layers.22.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #303

- └ Name: base_model.layers.22.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #306

- └ Name: base_model.layers.23.self_attn.q_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=True)
- └ Parameters: 803,712
- └ Status: NON-TRAINABLE

Layer #307

- └ Name: base_model.layers.23.self_attn.k_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #308

- └ Name: base_model.layers.23.self_attn.v_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=128, bias=True)
- └ Parameters: 114,816
- └ Status: NON-TRAINABLE

Layer #309

- └ Name: base_model.layers.23.self_attn.o_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=896, bias=False)
- └ Parameters: 802,816
- └ Status: NON-TRAINABLE

Layer #311

- └ Name: base_model.layers.23.mlp.gate_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #312

- └ Name: base_model.layers.23.mlp.up_proj
- └ Type: Linear
- └ Details: Linear(in_features=896, out_features=4864, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #313

- └ Name: base_model.layers.23.mlp.down_proj
- └ Type: Linear
- └ Details: Linear(in_features=4864, out_features=896, bias=False)
- └ Parameters: 4,358,144
- └ Status: NON-TRAINABLE

Layer #314

- └ Name: base_model.layers.23.mlp.act_fn
- └ Type: SiLU
- └ Details: SiLU()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #315

- └ Name: base_model.layers.23.input_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #316

- └ Name: base_model.layers.23.post_attention_layernorm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #317

- └ Name: base_model.norm
- └ Type: Qwen2RMSNorm
- └ Details: Qwen2RMSNorm((896,), eps=1e-06)
- └ Parameters: 896
- └ Status: NON-TRAINABLE

Layer #318

- └ Name: base_model.rotary_emb
- └ Type: Qwen2RotaryEmbedding
- └ Details: Qwen2RotaryEmbedding()
- └ Parameters: 0
- └ Status: NON-TRAINABLE

Layer #320

- | Name: attention_pooler.0
- | Type: Linear
- | Details: Linear(in_features=896, out_features=128, bias=True)
- | Parameters: 114,816
- | Status: TRAINABLE

Layer #321

- | Name: attention_pooler.1
- | Type: Tanh
- | Details: Tanh()
- | Parameters: 0
- | Status: NON-TRAINABLE

Layer #322

- | Name: attention_pooler.2
- | Type: Linear
- | Details: Linear(in_features=128, out_features=1, bias=True)
- | Parameters: 129
- | Status: TRAINABLE

Layer #324

- | Name: classifier.0
- | Type: Linear
- | Details: Linear(in_features=896, out_features=256, bias=True)
- | Parameters: 229,632
- | Status: TRAINABLE

Layer #325

- | Name: classifier.1
- | Type: LayerNorm
- | Details: LayerNorm((256,), eps=1e-05, elementwise_affine=True)
- | Parameters: 512
- | Status: TRAINABLE

Layer #326

- | Name: classifier.2
- | Type: GELU
- | Details: GELU(approximate='none')
- | Parameters: 0
- | Status: NON-TRAINABLE

Layer #327

- | Name: classifier.3
- | Type: Dropout
- | Details: Dropout(p=0.2, inplace=False)
- | Parameters: 0
- | Status: NON-TRAINABLE

Layer #328

- | Name: classifier.4

```
| Type: Linear
| Details: Linear(in_features=256, out_features=128, bias=True)
| Parameters: 32,896
| Status: TRAINABLE
```

Layer #329

```
| Name: classifier.5
| Type: LayerNorm
| Details: LayerNorm((128,), eps=1e-05, elementwise_affine=True)
| Parameters: 256
| Status: TRAINABLE
```

Layer #330

```
| Name: classifier.6
| Type: GELU
| Details: GELU(approximate='none')
| Parameters: 0
| Status: NON-TRAINABLE
```

Layer #331

```
| Name: classifier.7
| Type: Dropout
| Details: Dropout(p=0.1, inplace=False)
| Parameters: 0
| Status: NON-TRAINABLE
```

Layer #332

```
| Name: classifier.8
| Type: Linear
| Details: Linear(in_features=128, out_features=2, bias=True)
| Parameters: 258
| Status: TRAINABLE
```

```
=====
=====
Total Trainable Parameters: 378,499
Total Non-Trainable Parameters: 494,032,768
Total Parameters: 494,411,267
Trainable Parameters Percentage: 0.08%
=====
=====
```

Use EOS token as padding

```
tokenizer.pad_token
```

```
{"type": "string"}
```

```
tokenizer.eos_token
```



```
{"type": "string"}
```

```
model.pad_token_id = tokenizer.eos_token_id # Use EOS token as padding
```

Train

```
class CustomTrainer(Trainer):
    def save_model(self, output_dir=None, _internal_call=False):
        """ 保存分類器權重和配置 """
        # 使用預設 output_dir 如果未指定
        output_dir = output_dir if output_dir is not None else
self.args.output_dir
        os.makedirs(output_dir, exist_ok=True)

        # 保存分類器權重
        classifier_path = os.path.join(output_dir,
"classifier_weights.pt")
        model_dict = {
            'classifier': self.model.classifier.state_dict(),
            'attention_pooler':
self.model.attention_pooler.state_dict(),
            'layer_weights': self.model.layer_weights,
            'config': {
                'num_labels': self.model.config.num_labels,
                'hidden_size': self.model.config.hidden_size
            }
        }

        # Save model config (required by HF Trainer)
        if hasattr(self.model, "config"):
            self.model.config.save_pretrained(output_dir)

        torch.save(model_dict, classifier_path)
        print(f"已保存分類器權重至 {classifier_path}")

        # 保存訓練狀態
        super().save_state()

        return output_dir

#metric = evaluate.combine(["accuracy", "f1", "precision", "recall"])
#metric = evaluate.combine(["accuracy"])
def compute_metrics(eval_pred):
    predictions, labels = eval_pred
    predictions = np.argmax(predictions, axis=1) # Convert
probabilities to predicted labels
```

```

metric = evaluate.load('accuracy')
return metric.compute(predictions=predictions, references=labels)

# requires several GB of GPU memory
training_args = TrainingArguments(
    output_dir="checkpoints_v4", # Output directory for checkpoints
    learning_rate=5e-5, # Learning rate for the optimizer
    weight_decay=0.01, # Weight decay for regularization
    warmup_steps=500,
    seed=101,

    per_device_train_batch_size=26, # Batch size per device
    per_device_eval_batch_size=3, # Batch size per device for
evaluation

    num_train_epochs=5, # Number of training epochs

    eval_strategy='steps', # Evaluate after each epoch
    save_strategy="steps", # Save model checkpoints after each epoch

    #load_best_model_at_end=True, # Load the best model based on the
chosen metric
    save_total_limit=2,
    push_to_hub=False, # Disable pushing the model to the Hugging
Face Hub
    report_to="none", # Disable logging to Weight&Bias
    #fp16=True, # 是否用此精度訓練Whether to use fp16 16-bit (mixed)
precision training instead of 32-bit training.
    #bf16=True, # for 新型GPU 才能設定bf16
    logging_steps=1000,
)

trainer = CustomTrainer(
    model=model, # The LoRA-adapted model
    #tokenizer=tokenizer, # Tokenizer for the model
    processing_class=tokenizer, # Tokenizer for the model

    train_dataset=train_data, # Training dataset
    eval_dataset=test_data, # Evaluation dataset
    args=training_args, # Training arguments
    compute_metrics=compute_metrics, # Function to calculate
evaluation metrics
)

```

Resume training from the last checkpoint if available

接續訓練

訓練後，手動載入之前訓練的分類器權重

```
# checkpoint_path = "./checkpoints_v3/checkpoint-3167"
# model_path = "trained_classifier_v3"
# model.load_model(checkpoint_path)
```

Let's train the model

```
%%time
model.config.use_cache = False # Disable cache for faster training
trainer.train()
# trainer.train(resume_from_checkpoint=True) # Resume training from a
# checkpoint
#trainer.train(resume_from_checkpoint="./checkpoints_v1/checkpoint-
#19002")
```

<IPython.core.display.HTML object>

已保存分類器權重至 checkpoints_v4/checkpoint-500/classifier_weights.pt

```
{"model_id": "fb34573132554e6aa48670e192d8ab39", "version_major": 2, "version_minor": 0}
```

已保存分類器權重至 checkpoints_v4/checkpoint-1000/classifier_weights.pt
已保存分類器權重至 checkpoints_v4/checkpoint-1500/classifier_weights.pt
已保存分類器權重至 checkpoints_v4/checkpoint-1830/classifier_weights.pt
CPU times: user 40min 22s, sys: 38.6 s, total: 41min 1s
Wall time: 41min 27s

TrainOutput(global_step=1830, training_loss=0.33586385758196724, metrics={'train_runtime': 2487.2269, 'train_samples_per_second': 19.098, 'train_steps_per_second': 0.736, 'total_flos': 0.0, 'train_loss': 0.33586385758196724, 'epoch': 5.0})

Make a test

```
text = "今天天氣真好，我很開心"
predict_sentiment(text, model, tokenizer, device)
```

```
{'text': '今天天氣真好，我很開心',
 'sentiment': '正面',
 'confidence': 0.97,
 'probabilities': {'負面': 0.03, '正面': 0.97}}
```

```
text = "這個產品品質差，服務更糟糕"
predict_sentiment(text, model, tokenizer, device)
```

```
{'text': '這個產品品質差，服務更糟糕',  
  'sentiment': '負面',  
  'confidence': 0.95,  
  'probabilities': {'負面': 0.95, '正面': 0.05}}
```

Save Model

```
# 存 lora_model  
# model_path = "trained_lora_model_v1"  
# trainer.model.save_pretrained(model_path)  
  
# 這樣存檔 - 只會儲存分類器權重和設定  
# model_path = "trained_classifier_v4"  
# trainer.save_model(model_path)  
# model.save_pretrained(model_path)  
  
# 這樣存檔 - 只會儲存分類器權重和設定  
model_path = "trained_classifier_v4"  
model.save_model(model_path)  
# model.save_pretrained(model_path)  
  
已保存分類器權重至 trained_classifier_v4/classifier_weights.pt
```

Evaluate the model

```
def predict(input_text):  
    inputs = tokenizer(input_text, return_tensors="pt").to(device) #  
    Convert to PyTorch tensors and move to GPU (if available)  
    with torch.no_grad():  
        #outputs = model(**inputs).logits # Get the model's output  
    logits  
        outputs = model(**inputs)['logits'] # Get the model's output  
    logits  
        y_prob = torch.sigmoid(outputs).tolist()[0] # Apply sigmoid  
    activation and convert to list  
    return np.round(y_prob, 2) # Round the predicted probability to 2  
    decimal places  
  
def inference(text):  
    inputs = tokenizer(text, return_tensors="pt", truncation=True,  
max_length=512).to(device)  
    with torch.no_grad():  
        outputs = model(**inputs)  
        probs = torch.softmax(outputs['logits'], dim=1)  
        #probs = torch.softmax(outputs.logits, dim=1)  
    return probs.argmax().item()
```

```
df_test = pd.DataFrame(data=test_data)
```

df_test

[illegible]

```
\n"description\\": \\\"\\\"\\n      }\\n      }\\n    ]\\n}\\", "type": "dataframe", "variable_name": "df_test"}
```

```
input_text='''立法院今天發布人事通報，總務處長周傑調任參事室參事並代理總務處長，引發外界聯想立法院長韓國瑜出手進行人事調整。周傑今天受訪表示，正常調動，謝謝關心。部分媒體報導，質疑立法院總務處長人事異動，是否與先前立法院立委辦公室傳出有裝潢費昂貴、不合理等情形，或是日前立法院司法及法制委員會有委員提議要更改格局，而周傑回應「建議要更動是否可在會期中研議規畫，休會時再改」，令國民黨立法院黨團總召傅崐萁不以為然有關。'''
```

```
predict(input_text)
```

```
array([0.31, 0.63])
```

```
input_text = "我不高興"
```

```
inference(input_text)
```

```
1
```

```
%%time
```

```
df_test['prediction'] = df_test['text'].map(predict)
```

```
df_test['y_pred'] = df_test['prediction'].apply(lambda x: np.argmax(x, axis=0))
```

```
#
```

```
# df_test['prediction'] = df_test['text'].map(inference)
```

```
CPU times: user 16.6 s, sys: 66.8 ms, total: 16.6 s
```

```
Wall time: 16.8 s
```

```
accuracy = (df_test['y_pred'] == df_test['label']).mean()
```

```
print(f"Model Accuracy on Test Data: {accuracy:.4f}")
```

```
df_test.head()
```

```
Model Accuracy on Test Data: 0.8980
```

```
{"summary": "{\\n  \\\"name\\\": \\\"df_test\\\",\\n  \\\"rows\\\": 500,\\n  \\\"fields\\\": [\\n    {\\n      \\\"column\\\": \\\"text\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"string\\\",\\n        \\\"num_unique_values\\\": 500,\\n        \\\"samples\\\": [\\n          \\\"\\\"\\n          u5751\\\"\\\"u6211\\\"\\\"u5427\\\"\\\"uff0c\\\"\\\"u8207\\\"\\\"u5716\\\"\\\"u7247\\\"\\\"u5b8c\\\"\\\"u5168\\\"\\\"u4e0d\\\"\\\"\\n          u7b26\\\"\\\"uff0c\\\"\\\"u5c0f\\\"\\\"u5f97\\\"\\\"u4e0d\\\"\\\"u884c\\\"\\\"uff0c\\\"\\\"u50cf\\\"\\\"u8ce3\\\"\\\"u4e0d\\\"\\\"\\n          u6389\\\"\\\"u5269\\\"\\\"u4e0b\\\"\\\"u7684\\\"\\\",\\n          \\\"\\\"\\\"u9019\\\"\\\"u500b\\\"\\\"u5982\\\"\\\"\\n          u5bb6\\\"\\\"u7b97\\\"\\\"u662f\\\"\\\"u96e2S0H0\\\"\\\"u73fe\\\"\\\"u4ee3\\\"\\\"u57ce\\\"\\\"u6700\\\"\\\"u8fd1\\\"\\\"\\n          u7684\\\"\\\"u4e00\\\"\\\"u500b\\\"\\\"u4e86,\\\"\\\"u6253\\\"\\\"u8eca\\\"\\\"u4e00\\\"\\\"u500b\\\"\\\"u8d77\\\"\\\"\\n          u6b65\\\"\\\"u5c31\\\"\\\"u53ef\\\"\\\"u4ee5\\\"\\\"u5230S0H0\\\"\\\"u73fe\\\"\\\"u4ee3\\\"\\\"u57ce\\\"\\\"u7684\\\"\\\"\\n          u5730\\\"\\\"u9435.\\\"\\\"u51fa\\\"\\\"u5dee\\\"\\\"u53bb\\\"\\\"u4f4f\\\"\\\"u4e868\\\"\\\"u5929.\\\"\\\"u9810\\\"\\\"\\n          u5b9a\\\"\\\"u7684\\\"\\\"u6642\\\"\\\"u5019\\\"\\\"u8ddf\\\"\\\"u651c\\\"\\\"u7a0b\\\"\\\"u7684\\\"\\\"u5de5\\\"\\\"u4f5c\\\"\\\"\\n          u4eba\\\"\\\"u54e1\\\"\\\"u8aaa\\\"\\\"u4e0d\\\"\\\"u8981\\\"\\\"u9760\\\"\\\"u8fd1\\\"\\\"u96fb\\\"\\\"u68af\\\"\\\"u4e0d\\\"\\\"\\n          u8981\\\"\\\"u8d70\\\"\\\"u5eca\\\"\\\"u76e1\\\"\\\"u982d\\\"\\\"u7684,\\\"\\\"u5165\\\"\\\"u4f4f\\\"\\\"u7684\\\"\\\"\\n          u6642\\\"\\\"u5019\\\"\\\"u679c\\\"\\\"u7136\\\"\\\"u7d66\\\"\\\"u5b89\\\"\\\"u6392\\\"\\\"u59a5\\\"\\\"u7576.\\\"\\\"\\n        ]\\n      }\\n    ]\\n  }\\n}
```

```

u524d\\u53f0\\u670d\\u52d9\\u4eba\\u54e1\\u614b\\u5ea6\\u90fd\\u5f88\\
u89aa\\u5207.\\u4f4f\\u9032\\u53bb\\u767c\\u73fe\\u5bec\\u5e36\\
u4e0d\\u80fd\\u4e0a\\u7db2,\\u8981\\u6c42\\u63db\\u623f,\\u4f46\\
u662f\\u7576\\u5929\\u5df2\\u7d93\\u6c92\\u6709\\u7a7a\\u623f\\
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u4e86,\\u7092\\u83dc\\u4e5f\\u4e0d\\u932f,\\u5f88\\u5bb6\\u5e38\\
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u9593\\u88e1\\u6309\\u6469.\\u624b\\u6cd5\\u9084\\u7b97\\u904e\\
u7684\\u53bb.\\u6700\\u5f8c\\u5168\\u958b\\u4e86\\u9910\\u98f2\\
u767c\\u7968\\u88e1,\\u56de\\u4f86\\u53ef\\u4ee5\\u5831\\u92b7\\
u4e86\\u54c8\\u54c8...\\u7e3d\\u7684\\u4f86\\u8aaa,200\\u591a\\u584a\\
u9322\\u7684\\u8cd3\\u9928,\\u9019\\u500b\\u5982\\u5bb6\\u7b97\\
u662f\\u6027\\u50f9\\u6bd4\\u5f88\\u9ad8\\u7684\\u4e86.",\\n
\\"\\u8a2d\\u65bd\\u9673\\u820a\\uff0c\\u5c31\\u662f\\u56e0\\u70ba\\
u96e2\\u6a5f\\u5834\\u8fd1\\uff0c\\u6240\\u4ee5\\u624d\\u9078\\u64c7\\
u9019\\u88e1\\"\\n
    ],\\n
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    "description": \\"\\n    }\\n    },\\n
    {"\\n    "column":
    "label",\\n
    "properties": {\\n
    "dtype": "number",\\n
    "std": 0,\\n
    "min": 0,\\n
    "max": 1,\\n
    "num_unique_values": 2,\\n
    "samples": [\\n
    1,\\n
    0\\n
    ],\\n
    "semantic_type": \\"\\",\\n
    "description": \\"\\n    }\\n    },\\n
    {"\\n    "column":
    "input_ids",\\n
    "properties": {\\n
    "dtype":
    "object",\\n
    "semantic_type": \\"\\",\\n
    "description": \\"\\n    }\\n    },\\n
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    "properties": {\\n
    "dtype":
    "object",\\n
    "semantic_type": \\"\\",\\n
    "description": \\"\\n    }\\n    },\\n
    {"\\n    "column":
    "prediction",\\n
    "properties": {\\n
    "dtype":
    "object",\\n
    "semantic_type": \\"\\",\\n
    "description": \\"\\n    }\\n    },\\n
    {"\\n    "column":
    "y_pred",\\n
    "properties": {\\n
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    "std": 0,\\n
    "min": 0,\\n
    "max": 1,\\n
    "num_unique_values": 2,\\n
    "samples": [\\n
    1,\\n
    0\\n
    ],\\n
    "semantic_type": \\"\\",\\n
    "description": \\"\\n    }\\n    }\\n
    ]\\n
    }", "type": "dataframe", "variable_name": "df_test"}

```

```

cm = metrics.confusion_matrix(df_test['label'], df_test['y_pred'])
cm

array([[215,  31],
       [ 20, 234]])

```

```
print(metrics.classification_report(df_test['label'],
df_test['y_pred']))
```

	precision	recall	f1-score	support
0	0.91	0.87	0.89	246
1	0.88	0.92	0.90	254
accuracy			0.90	500
macro avg	0.90	0.90	0.90	500
weighted avg	0.90	0.90	0.90	500

```
metrics.precision_score(df_test['label'], df_test['y_pred'],
average='micro')
```

0.898

```
metrics.precision_score(df_test['label'], df_test['y_pred'],
average='macro')
```

0.8989562424729025

```
metrics.recall_score(df_test['label'], df_test['y_pred'],
average='micro')
```

0.898

```
metrics.recall_score(df_test['label'], df_test['y_pred'],
average='macro')
```

0.8976217911785418

Load model

```
# 在外部先載入base_model 預訓練權重(不包含分類層)
# full_model = AutoModelForCausalLM.from_pretrained(model_id)
#
model_path = "trained_classifier_v4"
# model_path = "checkpoints_v3/checkpoint-4145"
hidden_size = full_model.config.hidden_size
model = QwenForClassifier(full_model.model, hidden_size, num_labels=
len(categories))

model.load_model(model_path, device=device)
```



```
# 移動到指定設備
```

```
model = model.to(device)
```

```
已載入分類器權重: trained_classifier_v4/classifier_weights.pt
```

```
text = "這個產品品質差，服務更糟糕"
```

```
predict_sentiment(text, model, tokenizer, device)
```

```
{'text': '這個產品品質差，服務更糟糕',  
 'sentiment': '負面',  
 'confidence': 0.93,  
 'probabilities': {'負面': 0.93, '正面': 0.07}}
```

```
full_model
```

```
Qwen2ForCausalLM(  
  (model): Qwen2Model(  
    (embed_tokens): Embedding(151936, 896)  
    (layers): ModuleList(  
      (0-23): 24 x Qwen2DecoderLayer(  
        (self_attn): Qwen2Attention(  
          (q_proj): Linear(in_features=896, out_features=896,  
bias=True)  
          (k_proj): Linear(in_features=896, out_features=128,  
bias=True)  
          (v_proj): Linear(in_features=896, out_features=128,  
bias=True)  
          (o_proj): Linear(in_features=896, out_features=896,  
bias=False)  
        )  
        (mlp): Qwen2MLP(  
          (gate_proj): Linear(in_features=896, out_features=4864,  
bias=False)  
          (up_proj): Linear(in_features=896, out_features=4864,  
bias=False)  
          (down_proj): Linear(in_features=4864, out_features=896,  
bias=False)  
          (act_fn): SiLU()  
        )  
        (input_layernorm): Qwen2RMSNorm((896,), eps=1e-06)  
        (post_attention_layernorm): Qwen2RMSNorm((896,), eps=1e-06)  
      )  
    )  
    (norm): Qwen2RMSNorm((896,), eps=1e-06)  
    (rotary_emb): Qwen2RotaryEmbedding()  
  )  
  (lm_head): Linear(in_features=896, out_features=151936, bias=False)  
)
```

Text generation fro full_model

```
from IPython.display import Markdown

def generate_text(input_prompt):

    messages = [
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": input_prompt}
    ]
    text = tokenizer.apply_chat_template(
        messages,
        tokenize=False,
        add_generation_prompt=True
    )
    model_inputs = tokenizer([text], return_tensors="pt").to(device)

    prompt_length = model_inputs['input_ids'].shape[1]

    generated_ids = full_model.generate(
        model_inputs.input_ids,
        max_new_tokens=512,
        pad_token_id=tokenizer.eos_token_id
    )
    response = tokenizer.decode(generated_ids[0][prompt_length:],
    skip_special_tokens=True)
    return response
```

```
text="給出三個保持健康的提示。"
result = generate_text(text)
Markdown(result)
```

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results.

<IPython.core.display.Markdown object>

```
%%time
text="我們如何減少空氣污染？請給幾項重要的建議。"
result = generate_text(text)
Markdown(result)
```

CPU times: user 10.4 s, sys: 36.7 ms, total: 10.5 s
Wall time: 10.5 s

<IPython.core.display.Markdown object>